

UNIVERSITY OF SOUTHERN CALIFORNIA

# EE 483: Digital Signal Processing

## Lecture 8: The Discrete Fourier Transform (DFT)

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### 1 The Discrete Fourier Transform (DFT)

**Def.**

The DFT is a mapping of a length- $N$  sequence  $x[n]$ ,  $0 \leq n \leq N - 1$ , into a length- $N$  Fourier representation  $X[k]$ ,  $k = 0, \dots, N - 1$ , according to

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}. \quad (\text{Mitra 5.7})$$

It is a **sampled version of the DTFT** for a finite-length sequence:<sup>1</sup>

$$X[k] = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}.$$

$\Rightarrow$  discrete in both time and frequency.

**Inverse DFT.**

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j \frac{2\pi kn}{N}}, \quad n = 0, \dots, N - 1.$$

<sup>1</sup>See the DTFT definition in Lecture 5 (September 8):  $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$  (Mitra 3.10).

2

## 1.1 Twiddle-Factor Notation

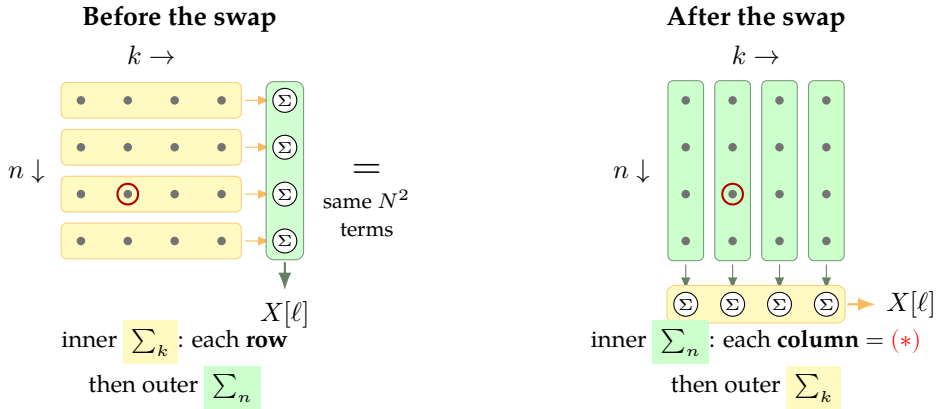
With  $W_N = e^{-j\frac{2\pi}{N}}$ ,

Rewrite the DFT:  $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1. \quad (\text{Mittra 11.60})$

Rewrite the Inverse DFT:  $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq n \leq N-1.$

**Verification of the inverse formula (check that the IDFT is indeed the inverse of the DFT).**

$$\begin{aligned}
 X[\ell] &= \sum_{n=0}^{N-1} x[n] W_N^{\ell n} \\
 &= \sum_{n=0}^{N-1} \left[ \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \right] W_N^{\ell n} \\
 &= \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n} \quad (\text{move all factors into } \sum_k) \\
 &= \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n} \quad (\text{swap the sums}) \\
 &= \sum_{k=0}^{N-1} X[k] \underbrace{\left[ \frac{1}{N} \sum_{n=0}^{N-1} W_N^{-(k-\ell)n} \right]}_{(*)}.
 \end{aligned}$$



Each dot is one term  $a[n, k] = \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n}$  (drawn for  $N = 4$ ; illustration, not in orig lecture note). The red ring is the same term  $a[2, 1]$  in both grids: it moves from “row 2” to “column 1” but is not changed.

Summing the geometric series (recall  $W_N = e^{-j\frac{2\pi}{N}}$ , so  $W_N^{-(k-\ell)} = e^{j\frac{2\pi(k-\ell)}{N}}$ ),

$$\begin{aligned}
 (*) &= \frac{1}{N} \frac{1 - W_N^{-(k-\ell)N}}{1 - W_N^{-(k-\ell)}} = \frac{1}{N} \frac{1 - e^{j2\pi(k-\ell)}}{1 - e^{j\frac{2\pi(k-\ell)}{N}}} \\
 &= \frac{1}{N} \frac{e^{j\pi(k-\ell)} (e^{-j\pi(k-\ell)} - e^{j\pi(k-\ell)})}{e^{j\frac{\pi(k-\ell)}{N}} (e^{-j\frac{\pi(k-\ell)}{N}} - e^{j\frac{\pi(k-\ell)}{N}})} \\
 &= \frac{1}{N} \frac{\sin(\pi[k - \ell]) e^{j\pi(k-\ell)}}{\sin(\frac{\pi}{N}[k - \ell]) e^{j\frac{\pi(k-\ell)}{N}}} \quad 2
 \end{aligned}$$

3

$$= \begin{cases} 1, & k \equiv \ell \pmod{N}, \\ 0, & \text{else} \end{cases} \quad 3$$

<sup>2</sup>The  $\sin / \sin$  factor is the *Dirichlet kernel*  $D_N(\omega) = \frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})}$  (Lecture 5, September 8, § 2), at  $\omega = \frac{2\pi(\mathbf{k} - \ell)}{N}$ .

In this verification,  $k, \ell \in \{0, \dots, N-1\}$ , so  $k \equiv \ell \pmod{N} \iff k = \ell$ .

$$= \delta[\langle k - \ell \rangle_N].$$

**Modulo Operation  $\langle n \rangle_N$ .**

$k = \langle n \rangle_N$  if  $k \in \{0, \dots, N-1\}$  and there exists an integer  $\ell$  such that  $n = k + \ell N$ .

**Example  $\langle n \rangle_3$**  (see Mitra 2.3.1).

| $n$                   | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|-----------------------|----|----|---|---|---|---|---|---|---|
| $\langle n \rangle_3$ | 1  | 2  | 0 | 1 | 2 | 0 | 1 | 2 | 0 |

Therefore

$$X[\ell] = \sum_{k=0}^{N-1} X[k] \delta[\langle k - \ell \rangle_N] = X[\ell] \quad \text{for } \ell \in \{0, \dots, N-1\}. \blacksquare$$

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<sup>3</sup>The geometric-series formula is only valid when  $W_N^{-(k-\ell)} \neq 1$ , i.e.,  $k \not\equiv \ell \pmod{N}$ ; then  $\sin(\pi[k - \ell]) = 0$  while  $\sin(\frac{\pi}{N}[k - \ell]) \neq 0$ , so  $(*) = 0$ . When  $k \equiv \ell \pmod{N}$ ,  $W_N^{-(k-\ell)} = 1$ , so every term  $W_N^{-(k-\ell)n} = 1$  and  $(*) = \frac{1}{N} \cdot N = 1$  directly (also the 0/0 limit of the Dirichlet-kernel expression).

4

**Example.**

$g[n] = \cos\left(\frac{2\pi rn}{N}\right)$ ,  $0 \leq n \leq N-1$ , with  $r$  an integer,  $0 \leq r \leq N-1$ . We can write  $g[n]$  as

$$\begin{aligned} g[n] &= \frac{1}{2} \left( e^{j\frac{2\pi rn}{N}} + e^{-j\frac{2\pi rn}{N}} \right) && \text{(Euler)} \\ &= \frac{1}{2} (W_N^{-rn} + W_N^{rn}) && \text{(recall } W_N = e^{-j\frac{2\pi}{N}}). \end{aligned}$$

$$\begin{aligned} G[k] &= \sum_{n=0}^{N-1} g[n] W_N^{kn} = \sum_{n=0}^{N-1} \frac{1}{2} (W_N^{-rn} + W_N^{rn}) W_N^{kn} \\ &= \frac{1}{2} \left( \sum_{n=0}^{N-1} W_N^{-rn} W_N^{kn} + \sum_{n=0}^{N-1} W_N^{rn} W_N^{kn} \right) \\ &= \frac{1}{2} \left( \sum_{n=0}^{N-1} W_N^{-(r-k)n} + \sum_{n=0}^{N-1} W_N^{-(r+k)n} \right) \\ &= \frac{1}{2} \left( \textcolor{red}{N} \delta[\langle r-k \rangle_N] + \textcolor{red}{N} \delta[\langle -r-k \rangle_N] \right). \end{aligned}$$

For  $0 \leq k \leq N-1$ :

$$G[k] = \begin{cases} \frac{N}{2}, & k = r, \quad k = N-r, \\ 0, & \text{else,} \end{cases} \quad \text{(unless } r = 0 \text{ or } r = N/2)^4$$

using  $\langle -r-k \rangle_N = \langle N-r-k \rangle_N$ . When  $r = 0$  or  $r = N/2$ , the two impulses coincide and  $G[r] = N$ .

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Reading the two  $\delta$ 's for  $k \in \{0, \dots, N-1\}$ :

- $k = r$ :  $\langle r-k \rangle_N = 0$  in the first term.
- $k = N-r$ :  $\langle -r-k \rangle_N = 0$  in the second term, i.e.  $k \equiv -r \pmod{N}$  (the representative in  $\{0, \dots, N-1\}$  is  $N-r$ ).

5

## 2 DFT Properties

5

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<sup>5</sup>cf. Mitra Tables 5.1, 5.2, and 5.3 (reproduced in the appendix of the full build).

## 2.1 Periodicity

### Property.

$$\begin{aligned} X[\ell N + k] &= X[k], & k = 0, \dots, N-1, \ell \text{ an integer,} \\ x[mN + n] &= x[n], & n = 0, \dots, N-1, m \text{ an integer,} \end{aligned}$$

when the DFT / IDFT formulas are evaluated outside  $0, \dots, N-1$ .

Because

$$\begin{aligned} W_N^N &= e^{-j\frac{2\pi}{N}N} = e^{-j2\pi} = 1 \\ \Rightarrow W_N^{\ell N + k} &= (1)^\ell W_N^k = W_N^k. \end{aligned}$$

So adding  $\ell N$  to a frequency index (or  $mN$  to a time index) does not change any twiddle factor in the DFT/IDFT sums. We therefore extend  $X[k]$  and  $x[n]$  periodically with period  $N$  when those formulas are used for integer indices outside  $\{0, \dots, N-1\}$  (as stated in the Property box).

## 2.2 Circular Shift

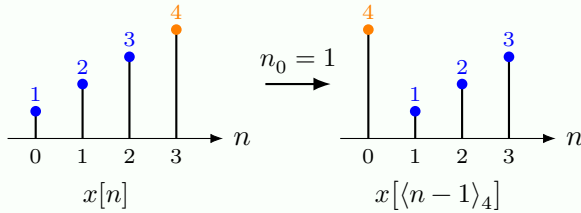
Consider  $x[n]$ ,  $0 \leq n \leq N-1$ . The ordinary shift  $x_1[n] = x[n - n_0]$  is **no longer defined** over  $0 \leq n \leq N-1$ .

$$\Rightarrow \text{circular shifts:} \quad x_1[n] = x[\langle n - n_0 \rangle_N]. \quad (\text{Mittra 2.25})$$

6

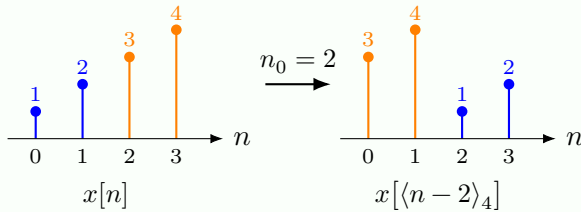
**Example (Right Circular Shift).**Let  $0 < n_0 \leq N - 1$ :

$$x_1[n] = x[\langle n - n_0 \rangle_N] = \begin{cases} x[n - n_0], & n_0 \leq n \leq N - 1, \\ x[N + n - n_0], & 0 \leq n \leq n_0 - 1. \end{cases} \quad (\text{Mitra 2.26})$$



$$x[n] = \{\underset{\uparrow}{1}, 2, 3, 4\}, \quad x[\langle n-1 \rangle_4] = \{\underset{\uparrow}{4}, 1, 2, 3\},$$

$$x[\langle n-2 \rangle_4] = \{\underset{\uparrow}{3}, 4, 1, 2\} = x[\langle n+2 \rangle_4].$$

**Property (Circular Time-Shifting).**Let  $g[n] \leftrightarrow G[k]$  and  $\tilde{g}[n] = g[\langle n - n_0 \rangle_N]$ . Then

$$\tilde{G}[k] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k].$$

7

**Proof.**

$$\tilde{G}[k] = \sum_{n=0}^{N-1} g[\langle n - n_0 \rangle_N] W_N^{kn} = \sum_{n=0}^{n_0-1} g[n + N - n_0] W_N^{kn} + \sum_{n=n_0}^{N-1} g[n - n_0] W_N^{kn}.$$

Let  $m = n + N - n_0$  ( $n = m - N + n_0$ ) in the first sum and  $p = n - n_0$  ( $n = p + n_0$ ) in the second:

$$\begin{aligned} \tilde{G}[k] &= \sum_{m=N-n_0}^{N-1} g[m] W_N^{k(m-N+n_0)} + \sum_{p=0}^{N-1-n_0} g[p] W_N^{k(p+n_0)} \\ &= \sum_{n=0}^{N-1} g[n] W_N^{k(n+n_0)} \quad (W_N^{-kN} = 1) \\ &= W_N^{kn_0} \left[ \sum_{n=0}^{N-1} g[n] W_N^{kn} \right] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k]. \quad \blacksquare \end{aligned}$$

8

## 2.3 Convolution

### 2.3.1 Linear convolution and the DTFT

$$x[n] \otimes h[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) H(e^{j\omega}).$$

If  $x[n]$  has length  $N_1$  and  $h[n]$  has length  $N_2$ , then  $y[n] = x[n] \otimes h[n]$  has length  $N_1 + N_2 - 1$ : it is possibly nonzero for  $0 \leq n \leq N_1 + N_2 - 2$  and is zero otherwise.

### 2.3.2 Circular Convolution (two length- $N$ sequences)

**Def.**

$$y_c[n] = \sum_{m=0}^{N-1} x[m] h[\langle n - m \rangle_N], \quad 0 \leq n \leq N-1, \quad (\text{Mitra 5.51})$$

$$y_c[n] = x[n] \circledast h[n] = h[n] \circledast x[n]. \quad (\text{Mitra 5.52})$$



**Convolution Theorem.**

If  $y_c[n] = x[n] \circledast h[n]$ , then  $Y_c[k] = X[k] H[k]$ .

**9****Proof.**

$$\begin{aligned}
 Y_c[k] &= \sum_{n=0}^{N-1} \left[ \sum_{m=0}^{N-1} x[\langle n-m \rangle_N] h[m] \right] W_N^{kn} \\
 &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x[\langle n-m \rangle_N] h[m] W_N^{kn} \quad (\text{expand inner sum}) \\
 &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} x[\langle n-m \rangle_N] h[m] W_N^{kn} \quad (\text{swap the sums}) \\
 &= \sum_{m=0}^{N-1} h[m] \left[ \sum_{n=0}^{N-1} x[\langle n-m \rangle_N] W_N^{kn} \right] \\
 &= \left[ \sum_{m=0}^{N-1} h[m] W_N^{km} \right] X[k] \quad (\text{circular time-shift}) \\
 &= H[k] X[k].
 \end{aligned}$$

■

**2.4 Linear Convolution via Circular Convolution**

If  $x[n], 0 \leq n \leq N_1 - 1$ , and  $h[n], 0 \leq n \leq N_2 - 1$ , then

$$y[n] = x[n] \otimes h[n], \quad 0 \leq n \leq N_1 + N_2 - 2.$$

**Property.**

If

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N_1 - 1, \\ 0, & N_1 \leq n < N, \end{cases} \quad \text{where } N \geq N_1 + N_2 - 1,$$

and similarly define a “zero-padded” length- $N$   $\tilde{h}[n]$ , then

$$\tilde{x}[n] \textcircled{N} \tilde{h}[n] = x[n] \otimes h[n] \quad \text{for } 0 \leq n \leq N - 1.$$

10

**Consequence: we can compute  $\otimes$  using  $\textcircled{N}!$ .**

1. Zero-pad both sequences to the same length  $N \geq N_1 + N_2 - 1$ .
2. Take length- $N$  DFTs of both.
3. Multiply the DFTs together.
4. Take the inverse DFT of the result.

**Remark***(addendum to orig lecture note).*

Why  $N \geq N_1 + N_2 - 1$ : circular convolution equals the linear convolution *aliased* with period  $N$ ,  $\tilde{x}[n] \textcircled{N} \tilde{h}[n] = \sum_r y[n + rN]$  for  $0 \leq n \leq N - 1$ . Since  $y[n]$  occupies  $N_1 + N_2 - 1$  samples, no shifted copies overlap exactly when  $N \geq N_1 + N_2 - 1$ . With a fast algorithm (FFT) for the DFT, this is how long convolutions are computed in practice (cf. Mitra Ch. 11).

**2.5 Parseval's Theorem****Property**

6.

$$\begin{aligned} \sum_{n=0}^{N-1} x^*[n] y[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] Y[k], \\ \Rightarrow \sum_{n=0}^{N-1} |x[n]|^2 &= \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2. \end{aligned} \quad (\text{Mitra 5.6})$$

The proof was asked in Problem Set 5, Problem 1.

## 2.6 Symmetry

### Property.

If  $x[n]$  is real,

$$X[k] = X^*[N - k] = X^*[\langle -k \rangle_N], \quad 0 \leq k \leq N - 1.$$

(For  $k = 0$ , read  $X^*[N]$  as  $X^*[0]$  by periodicity; the modulo form handles this automatically.)

## Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5.4, 5.5, 5.7, 11.3–11.5
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 6, 7

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<sup>6</sup>Notation vs. Mitra:

- Sec. 5.1 is for a general orthogonal transform with basis  $\psi[k, n]$ . Its coefficients are written  $\mathcal{X}[k]$  (calligraphic).  $X[k]$  is used only for the DFT, from Sec. 5.2 on.
- Sec. 5.1 has only one sequence. Mitra (5.6) is the energy form  $\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |\mathcal{X}[k]|^2$ .
- Its derivation puts the conjugate on the second factor:  $\sum_n x[n] x^*[n] = \frac{1}{N} \sum_k \mathcal{X}[k] \mathcal{X}^*[k]$ .
- The two-sequence line above is Mitra (5.115) in Sec. 5.7:  $\sum_n g[n] h^*[n] = \frac{1}{N} \sum_k G[k] H^*[k]$ .
- This lecture writes  $x, y$  instead of  $g, h$  and puts the conjugate on the first factor. Both sides are then the complex conjugates of Mitra's, so the equation still holds.